Five Axes of Fragility: A Systematic Robustness Audit of Gene-Regulatory-Network Inference from Single-Cell Foundation Models

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Abstract

Motivation: Gene-regulatory-network (GRN) inference from single-cell transcriptomics is increasingly used to prioritize regulatory hypotheses, yet most benchmarks evaluate accuracy under fixed preprocessing without testing whether rankings survive realistic analytical perturbations.

Results: We audit GRN edge rankings across five fragility axes—cell count, clustering resolution, rare-cell reliability, donor generalization, and integration sensitivity—on four Tabula Sapiens tissues and one external cohort. Unconstrained rankings require >3,000 cells for stability; dual-null calibration universally recommends coarse clustering; rare cell types achieve 0% reliability-gate passage versus 18% for abundant types; 6–8 donors are needed for immune generalization; and Harmony integration displaces 31–45% of top-500 edges without perturbation-confirmed improvement. No dataset passes all five axes, and corrective measures along one axis do not resolve others.

Availability and implementation: Code and pipelines are available at `https://github.com/Biodyn-AI/grn-robustness-audit`.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Single-cell RNA-sequencing has enabled transcriptome-wide inference of gene regulatory networks (GRNs), with attention-derived edge scores from foundation models such as scGPT (Cui et al., 2024) offering a scalable alternative to classical methods like SCENIC (Aibar et al., 2017) and GRNBoost2 (Moerman et al., 2019). These methods rank transcription-factor–target (TF–target) edges and interpret top-ranked pairs as regulatory hypotheses, but confidence in those rankings depends on analytical choices that are rarely stress-tested in practice.

Most published GRN benchmarks evaluate accuracy against a single reference database under default preprocessing parameters (Pratapa et al., 2020; Chen and Mar, 2018). This leaves open the question of whether the same edges would be prioritized if the analyst had used a different cell count, clustering resolution, integration method, or donor cohort. The practical consequence is that two groups analyzing the same atlas can reach different biological conclusions simply by making different preprocessing decisions, without either being demonstrably wrong.

We address this gap by conducting a systematic multi-axis robustness audit of attention-derived TF–target edge rankings. Our audit spans five axes that correspond to the most common sources of analytical variation in single-cell GRN workflows:

1. **Cell-count subsampling:** How many cells are needed for stable rankings, and do prior-constrained policies rescue small-sample regimes?

2. **Cluster-resolution sensitivity:** Does switching between coarse and fine clustering change which edges are prioritized?

3. **Rare-cell-type reliability:** Do rare cell types yield mechanistic edges that pass the same confidence gates as abundant types?

4. **Donor generalization:** How many donors are needed for rankings to transfer to held-out individuals?

5. **Integration-method sensitivity:** Does batch correction by Harmony alter edge rankings, and do the changes improve biological validity?

Each axis was developed through an adversarial-review loop in which an AI-assisted review agent systematically challenged the initial analysis for methodological weaknesses (e.g., confounded null designs, overfitted hyperparameters, hidden data leakage), corrective experiments were designed and executed, and claims were updated accordingly. The adversarial protocol is itself a contribution: it provides a reusable template for auditing inference pipelines before publication.

The remainder of this paper describes the shared experimental setup (Section 2), presents per-axis results (Section 3), synthesizes cross-axis interactions (Section 3.6), and provides a practical guidelines table for practitioners (Section 4).

2 Methods

2.1 Datasets and shared infrastructure

All analyses used processed single-cell datasets from the Tabula Sapiens atlas (The Tabula Sapiens Consortium, 2022) and one external lung cohort (Table 1). Datasets were preprocessed using Scanpy (Wolf et al., 2018) following established best practices (Luecken and Theis, 2019), with standard normalization, log-transformation, and highly-variable-gene selection. The same edge-scoring infrastructure was shared across all five axes: TF–target edges were scored using Pearson correlation on expression vectors, with TF panels derived from DoRothEA (Garcia-Alonso et al., 2019) or TRRUST (Han et al., 2018) depending on the axis.

Table 1: Datasets used across the five robustness axes.

Dataset	Cells	Donors	Batches	Axes
Kidney	8,000–11,376	1	2	1, 2, 5
Lung	15,000–20,000	4	7	2, 4, 5
Immune	15,000–20,000	12–20	42	2, 3, 4, 5
Immune inv.	15,000	—	—	2, 3
External lung	9,000	—	—	2, 3

2.2 Axis 1: Cell-count subsampling

We evaluated cell-count robustness on kidney tissue across three size tiers (small, medium, large) at 200, 1,000, and 3,000 cells over 22 completed runs. Five edge-ranking policies were compared: `unconstrained` (no prior filtering), `tf_source` and `tf_source_target` (DoRothEA-constrained), and matched random controls (`random_source`, `random_source_target`) where random gene sets of identical cardinality were drawn independently per run to avoid cross-run set leakage. The random-null ensemble comprised 64 independent draws, and empirical one-sided p-values with Benjamini–Hochberg (BH) correction were computed for prior-versus-random comparisons. The Minimum Valid Cell Count (MVCC) was defined as the smallest cell count at which unconstrained top-1,000 reference Jaccard exceeds 0.5 relative to the 3,000-cell anchor.

2.3 Axis 2: Cluster-resolution sensitivity

Coarse-versus-fine clustering sensitivity was assessed across five cohorts (kidney, lung, immune, immune-invariant, external lung). Coarse labels used `broad_cell_class` or `compartment`; fine labels used Leiden subclusters enforced as strict hierarchical refinements via `coarse:::fine` concatenation. Edge scores combined global and grouped maximum absolute Pearson correlation ($S = 0.6R_{\text{global}} + 0.4R_{\text{group_max}}$) over the top 600 high-variance genes. Agreement was quantified by a Resolution Sensitivity Score (RSS):

$$\text{RSS} = 0.40(1 - \text{overlap}) + 0.30(1 - \text{Jaccard}) + 0.30 \cdot \text{median_drift} \tag{1}$$

at top-$k = 1{,}000$, where median_drift is the median absolute rank change of edges appearing in both the coarse and fine top-k lists. Two null families were applied: global cell-label shuffle and within-coarse constrained shuffle (49 permutations each). A dual-null calibration rule downgraded the base RSS recommendation one level per failed null family (fine $\rightarrow$ hybrid $\rightarrow$ coarse). The same dual-null calibration was applied at each point of a 3×3 hyperparameter grid (`top_genes` $\in \{400, 600, 800\}$, `global_weight` $\in \{0.4, 0.6, 0.8\}$).

2.4 Axis 3: Rare-cell-type reliability

Rare-cell-type stress testing covered 88 eligible cell types (minimum 40 cells) across four datasets. Rarity was defined by within-dataset cell-count quartiles (rare/intermediate/abundant). For each cell type, 8 bootstrap replicates at 80% subsampling were drawn, and top-$k = 100$ edges were scored. Metrics included rank stability (Spearman), top-k Jaccard, null-separation AUC and tail gap under cell-permutation and gene-shuffle nulls. A cell type was deemed reliable if stability ≥ 0.75, top-k Jaccard ≥ 0.35, primary null AUC ≥ 0.65, primary tail gap > 0, and cell count ≥ 80. Adversarial extensions included a dual-null stress test (minimum AUC across cell-permutation and gene-shuffle nulls), a shared-panel sensitivity analysis (36 TRRUST-intersected edges fixed across all datasets), and an external-lung cohort as an out-of-sample check.

2.5 Axis 4: Donor generalization

Donor generalization was assessed on immune (12–20 donors, 300 cells/donor) and lung (4 donors, 800 cells/donor) tissues using a fixed panel of 76 TFs and 108 targets (8,208 directed edges). An initial complementary-split protocol, in which increasing the train set mechanically shrank the holdout set, was replaced by a fixed-holdout design (holdout $= 2$ donors held constant) after adversarial review identified confounding between train-size and holdout-size effects. Three corrective protocols were executed: (A) fixed-holdout top-12 immune (train counts 2/4/6/8/10, 30 splits each), (B) expanded top-20 immune including long-tail donors (train counts 4/8/12/16, 40 splits each), and (C) composition-matched controls using broad cell-class matching (top-8 donors, 62 cells/donor). Transfer was quantified by Spearman ρ and top-k overlap between train and holdout edge-score vectors.

2.6 Axis 5: Integration-method sensitivity

The impact of Harmony batch correction (Korsunsky et al., 2019) was assessed on immune, lung, and kidney datasets. Harmony was applied in PCA space (default $n_{\text{pcs}} = 50$) using three batch keys (`_scvi_batch`, `donor_id`, `method`). Sensitivity metrics included rank Spearman,

sign-flip rate, mean absolute rank shift, and top-k overlap. External validation used precision@500 against DoRothEA (AB, ABCD, ChIP-seq variants) and TRRUST references, with paired swap-based binomial significance tests and BH-FDR correction. Perturbation validation used external Dixit-derived perturbation tables (Dixit et al., 2016) for immune and lung. A hyperparameter sweep over $n_{\text{pcs}} \in \{30, 50, 80\}$ tested the robustness–performance tradeoff.

2.7 Cross-axis synthesis

To assess interactions among axes, we constructed a cross-axis fragility profile summarizing, for each axis and dataset, the severity of the robustness concern (pass/marginal/fail relative to that axis's calibrated threshold). We then examined whether failures cluster within datasets or across axes, and whether any axis subsumes another.

3 Results

3.1 Axis 1: Cell count determines ranking stability, prior constraints rescue small samples

Unconstrained edge rankings exhibited severe instability at practical cell counts. Top-1,000 reference Jaccard against the 3,000-cell anchor was 0.001–0.029 at 200 cells and 0.037–0.145 at 1,000 cells across three kidney size tiers, yielding a conservative MVCC of >3,000 (Figure 1A).

Under the adversarially corrected per-run random null with 64 draws, random-policy stability was near zero (ensemble means 0.002–0.008 at 200/1,000 cells), while DoRothEA-constrained policies remained high: `tf_source` and `tf_source_target` achieved Jaccard 0.733–0.759 at 200 cells and 0.861–0.872 at 1,000 cells. Prior-versus-random empirical tests were significant after BH correction for both stability ($p = 0.0154$, $q \approx 0.017$) and biological support (high-cell DoRothEA support 0.055–0.066 for constrained versus 0.002–0.003 for random; Figure 1B).

The conclusion, demonstrated here in kidney tissue and likely applicable to other tissues given comparable sparsity regimes, is that unconstrained GRN inference is unreliable at cell counts below the MVCC threshold, and that informative prior constraints provide a principled

remedy that survives adversarial null scrutiny.

3.2 Axis 2: Dual-null calibration recommends conservative coarse clustering universally

Raw RSS values ranged from 0.276 (immune-invariant) to 0.440 (immune), suggesting a spread from "fine-stable" to "hybrid" under RSS-only guidance (Figure 2A). However, dual-null calibration inverted these recommendations. Global-shuffle null separation was passed only by kidney and lung (empirical $p = 0.02$); within-coarse constrained-null separation was passed by no cohort. Requiring dual-null agreement, all five datasets received a final calibrated recommendation of **coarse** (Figure 2B), including immune-invariant and external lung, which had appeared fine-stable by RSS alone.

Across the 3×3 hyperparameter sweep, dual-null pass rate was 0.00 for all datasets, with calibrated-recommendation consistency ranging from 0.67 (kidney) to 1.00 (immune, immune-invariant, external lung). RSS coefficient of variation across grid points was 0.30–0.41, confirming that raw sensitivity metrics are hyperparameter-dependent while dual-null-calibrated conclusions are robust.

Within-coarse heterogeneity remained substantial (mean within-class RSS: kidney 0.427, lung 0.346, immune 0.414, immune-invariant 0.244, external lung 0.394), with specific compartments such as kidney T cells (RSS 0.538) and immune T cells (RSS 0.497) exhibiting high sensitivity even under coarse-level pooling, which suggests that per-compartment heterogeneity should be reported alongside the coarse-level recommendation.

3.3 Axis 3: Rare cell types exhibit a systematic reliability gap

Across 88 cell types in four datasets, rare types were not significantly worse on global rank stability (Spearman: rare 0.884 vs. abundant 0.851, $p = 0.835$) but were significantly worse on decision-level metrics: top-k Jaccard (0.625 vs. 0.731, $p = 4.1 \times 10^{-3}$), null-separation AUC (0.520 vs. 0.620, $p = 5.3 \times 10^{-8}$), and null-tail gap (-0.062 vs. 0.082, $p = 1.5 \times 10^{-7}$). The reliability gate was passed by 0/22 rare cell types and 4/22 abundant cell types (Figure 3A–B).

Under the conservative dual-null (minimum AUC across cell-permutation and gene-shuffle models), the rare–abundant gap attenuated but remained significant (AUC gap 0.018, $p = 6.8 \times 10^{-4}$; tail gap 0.037, $p = 0.015$). Shared-panel sensitivity analysis showed heterogeneous agreement with adaptive-panel results (dataset Spearman 0.09–0.93 for null-AUC ordering): strong in immune and kidney but weak in lung, indicating that the rare-type deficit is partially but not fully explained by panel-construction choices.

Minimum reliable cell-size estimates ranged from 287 (external lung, model-based) to 16,486 (lung, empirical reliable minimum), underscoring that some tissues require very large cell-type samples before mechanistic edges reach reliability thresholds.

3.4 Axis 4: Donor generalization requires 6–8 donors and is partly composition-mediated

Under the corrected fixed-holdout protocol, immune-tissue transfer improved monotonically from train= 2 to train= 10 donors (Spearman 0.568 $\rightarrow$ 0.683; top-100 overlap 0.445 $\rightarrow$ 0.540), eliminating the apparent interior optimum from the initial complementary-split protocol (Figure 4A). The baseline complementary-split design had inflated transfer estimates by a mean of +0.080 Spearman and +0.102 top-100 overlap at shared train counts.

Expanding to top-20 donors (including long-tail low-count donors) lowered all transfer metrics (train= 16: Spearman 0.627, overlap@100 0.460), indicating that top-donor-only evaluation was optimistic for broad-cohort deployment. Updated recommendation using a 90%-of-best criterion is **6 train donors**; the 95%-of-best conservative criterion yields **8 train donors** (Figure 4B).

Composition-matched controls (top-8 donors, 3 shared broad classes, 62 cells/donor) reduced transfer estimates by 16.3% (Spearman: 0.707 $\rightarrow$ 0.591) and 10.7% (overlap@100: 0.610 $\rightarrow$ 0.545), confirming that cell-type composition explains a substantial but incomplete fraction of apparent donor generalization. Above-null signal remained robust even under the strictest protocol (lift@100, defined as observed overlap at rank 100 divided by chance expectation: 37.8–44.3$\times$).

3.5 Axis 5: Integration reorders rankings substantially but does not improve perturbation-grounded validity

Harmony integration produced moderate-to-large rank perturbations across all three tissues: Spearman correlation with baseline ranged from 0.757 to 0.823, sign-flip rates from 8.1% to 12.7%, and top-500 overlap from 0.552 to 0.688 (Figure 5A). Bootstrap confidence intervals confirmed stable effect sizes.

Curated-reference precision@500 showed nominal DoRothEA/TRRUST gains in immune and kidney, but after paired swap-based significance testing with BH-FDR correction, robust gains were concentrated in TRRUST only (immune and kidney: Δ precision +0.026 to +0.028, FDR < 0.05; Figure 5B). DoRothEA and ChIP-seq-backed gains were not FDR-significant. Perturbation validation using external Dixit-derived tables showed no significant Harmony improvement: top-500 perturbation swap deltas were small and negative (-0.002 to -0.004; FDR $= 1.0$).

A hyperparameter sweep over $n_{\text{pcs}} \in \{30, 50, 80\}$ revealed a robustness–performance trade-off: top-500 overlap increased monotonically with PCs (immune: $0.590 \rightarrow 0.736$), while DoRothEA precision peaked at intermediate PCs and declined at higher dimensionality.

3.6 Cross-axis fragility synthesis

Table 2 summarizes the fragility profile across all five axes and datasets, and Figure 6 visualizes the per-axis pass/marginal/fail status for each dataset. Several cross-axis interactions emerge:

- **Cell count constrains rare-type power**: the MVCC of >3,000 cells for unconstrained inference implies that most rare cell types (typically < 500 cells) are inherently underpowered, consistent with the 0% reliability-gate passage observed in Axis 3.
- **Resolution choice modulates donor transfer**: coarse clustering pools cell-type variation that would otherwise require per-donor composition matching, partially explaining why coarse-level recommendations are more robust.
- **Integration rotates shared covariance geometry**: Harmony's rank perturbation affects

the correlation structure on which cell-count stability, resolution sensitivity, and rare-type null separation all depend.

- **No axis is universally benign**: every dataset fails at least two axes by their respective calibrated criteria, and no single corrective (more cells, coarser clustering, more donors, integration) resolves all fragilities simultaneously.

Table 2: Per-axis robustness summary with key metric, calibrated threshold, and practical recommendation.

Axis	Key Metric	Threshold	Recommendation
Cell count	MVCC (top-1k Jaccard > 0.5)	$>$3,000 cells	Use $\geq$3,000 cells or apply prior constraints
Resolution	Dual-null pass rate	Both nulls must pass	Default to coarse; report within-class heterogeneity
Rare types	Reliability gate (5 criteria)	All must pass	Treat rare-cell edges as hypothesis-generating only
Donor	Fixed-holdout Spearman	90% of best	Collect $\geq$6 donors (8 conservative)
Integration	Perturbation swap Δ	FDR < 0.05	Report multi-reference validation; avoid universal claims

4 Discussion

This audit demonstrates that GRN edge rankings from single-cell data are fragile along multiple axes simultaneously, and that apparent robustness along one axis (e.g., coarse rank stability for rare types) can mask failure along another (e.g., null separation). The adversarial-review methodology that we applied to each axis is itself a contribution: by requiring corrective experiments before claims are updated, it systematically closes the gap between initial findings and defensible conclusions.

Practical implications. Table 3 provides concrete guidance for practitioners. The central message is that no single parameter choice can guarantee robust GRN inference; instead, robustness must be demonstrated along each relevant axis for the specific tissue, cell count, and analytical choices at hand.

Table 3: Practical guidelines for robust GRN inference from single-cell data.

Decision Point	Guideline
Sample size	Target $\geq$3,000 cells for unconstrained inference, or use informative prior constraints (DoRothEA/TRRUST) which stabilize rankings at 200+ cells.
Clustering	Default to coarse clustering unless dual-null calibration (global + within-coarse constrained shuffle) confirms that fine resolution adds separable signal. Report within-class RSS for compartment-level uncertainty.
Rare cell types	Report null-separation AUC alongside stability for every cell type. Flag rare types ($<$ 250 cells) as hypothesis-generating. Provide minimum-reliable-size estimates with uncertainty status.
Donor coverage	Collect $\geq$6 donors (8 for conservative generalization). Use fixed-holdout evaluation, not complementary splits. Report composition-controlled transfer alongside raw transfer.
Integration	Report at least paired swap significance for reference gains, plus one perturbation-backed metric when available. Include hyperparameter sensitivity (n_{pcs}). Avoid universal claims of biological improvement.
Cross-axis audit	Report a fragility profile (Table 2) covering all axes relevant to the study design. Acknowledge which axes were not testable and why.

Comparison to existing benchmarks. Published GRN benchmarks (Pratapa et al., 2020; Chen and Mar, 2018) have focused primarily on accuracy relative to gold-standard references, treating preprocessing as fixed. Our audit complements these by asking whether rankings are *stable* across realistic analytical perturbations, a necessary condition for practical utility. The two perspectives are not redundant: a method can be accurate under one configuration and fragile under perturbation, or robustly mediocre across all configurations.

Limitations. Several limitations apply across the audit. First, all edge scoring is correlation-based and does not directly estimate causal regulatory effects; although correlation on normalized counts avoids zero-inflation artifacts (Svensson, 2020), it remains susceptible to confounding by shared upstream regulators (Kharchenko et al., 2014). Second, the audit covers Tabula Sapiens tissues and one external lung cohort; generalization to disease tissues, developmental atlases, or cross-species comparisons remains untested. Third, adversarial null designs (particularly constrained shuffles) may be overly strict for certain scoring families, potentially leading to conservative recommendations that reject genuine fine-scale signal. Fourth, perturbation-backed validation data remain sparse, limiting statistical power for perturbation swap tests.

Fifth, we evaluated only Harmony among integration methods; broader comparisons including scVI and Scanorama (Gayoso et al., 2022; Hie et al., 2019), guided by recent integration benchmarks (Luecken et al., 2022; Tran et al., 2020), are warranted.

5 Conclusion

We present the first systematic multi-axis robustness audit of GRN inference from single-cell transcriptomics, spanning cell count, clustering resolution, rare-cell reliability, donor generalization, and integration-method sensitivity. The audit reveals that fragility is pervasive, axis-specific, and interactive: no dataset passed all five axes under calibrated criteria, and corrective measures along one axis do not automatically resolve others. We provide a practical guidelines table and an open-source framework, consisting of modular Python scripts for each axis (subsampling sweeps, resolution calibration with dual nulls, rare-type reliability gates, donor-holdout curves, and integration swap tests), that enables practitioners to conduct analogous audits for their own datasets and inference pipelines before publishing GRN claims.

Author Contributions

I.K. conceived the study, designed the five-axis robustness audit framework, implemented all computational pipelines, performed all statistical analyses, generated all figures and tables, and wrote the manuscript.

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Conflict of Interest

None declared.

Data and Code Availability

All source code, analysis pipelines, result tables, and figure-generation scripts are publicly available at https://github.com/Biodyn-AI/grn-robustness-audit. Raw single-cell data are from the Tabula Sapiens Consortium and are available at https://tabula-sapiens-portal.ds.czbiohub.org/.

Supplementary Information

Supplementary tables with per-dataset, per-axis detailed metrics and the full hyperparameter-grid results are available in the GitHub repository.

Abbreviations

AUC: Area Under the Curve; BH: Benjamini–Hochberg; FDR: False Discovery Rate; GRN: Gene Regulatory Network; MVCC: Minimum Valid Cell Count; PCA: Principal Component Analysis; RSS: Resolution Sensitivity Score; scRNA-seq: Single-Cell RNA Sequencing; TF: Transcription Factor.

References

Aibar, S., González-Blas, C. B., Moerman, T., et al. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nature Methods*, 14(11):1083–1086.

Chen, S. and Mar, J. C. (2018). Evaluating methods of inferring gene regulatory networks highlights their lack of performance for single cell gene expression data. *BMC Bioinformatics*, 19(1):232.

Cui, H., Wang, C., Maan, H., et al. (2024). scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480.

Dixit, A., Parnas, O., Li, B., et al. (2016). Perturb-Seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell*, 167(7):1853–1866.

Garcia-Alonso, L., Holland, C. H., Ibrahim, M. M., Turei, D., and Saez-Rodriguez, J. (2019). Benchmark and integration of resources for the estimation of human transcription factor activities. *Genome Research*, 29(8):1363–1375.

Gayoso, A., Lopez, R., Xing, G., et al. (2022). A Python library for probabilistic analysis of single-cell omics data. *Nature Biotechnology*, 40(2):163–166.

Han, H., Cho, J.-W., Lee, S., et al. (2018). TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Research*, 46(D1):D199–D202.

Hie, B., Bryson, B., and Berger, B. (2019). Efficient integration of heterogeneous single-cell transcriptomes using Scanorama. *Nature Biotechnology*, 37(6):685–691.

Kharchenko, P. V., Silberstein, L., and Scadden, D. T. (2014). Bayesian approach to single-cell differential expression analysis. *Nature Methods*, 11(7):740–742.

Korsunsky, I., Millard, N., Fan, J., et al. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296.

Luecken, M. D. and Theis, F. J. (2019). Current best practices in single-cell RNA-seq analysis: a tutorial. *Molecular Systems Biology*, 15(6):e8746.

Luecken, M. D., Büttner, M., Chaichoompu, K., et al. (2022). Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19(1):41–50.

Moerman, T., Aibar Santos, S., González-Blas, C. B., et al. (2019). GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks. *Bioinformatics*, 35(12):2159–2161.

Pratapa, A., Jalihal, A. P., Law, J. N., Bharadwaj, A., and Murali, T. M. (2020). Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 17(2):147–154.

Svensson, V. (2020). Droplet scRNA-seq is not zero-inflated. *Nature Biotechnology*, 38(2):147–150.

The Tabula Sapiens Consortium. (2022). The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896.

Tran, H. T. N., Ang, K. S., Chevrier, M., et al. (2020). A benchmark of batch-effect correction methods for single-cell RNA sequencing data. *Genome Biology*, 21(1):12.

Wolf, F. A., Angerer, P., and Theis, F. J. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1):15.

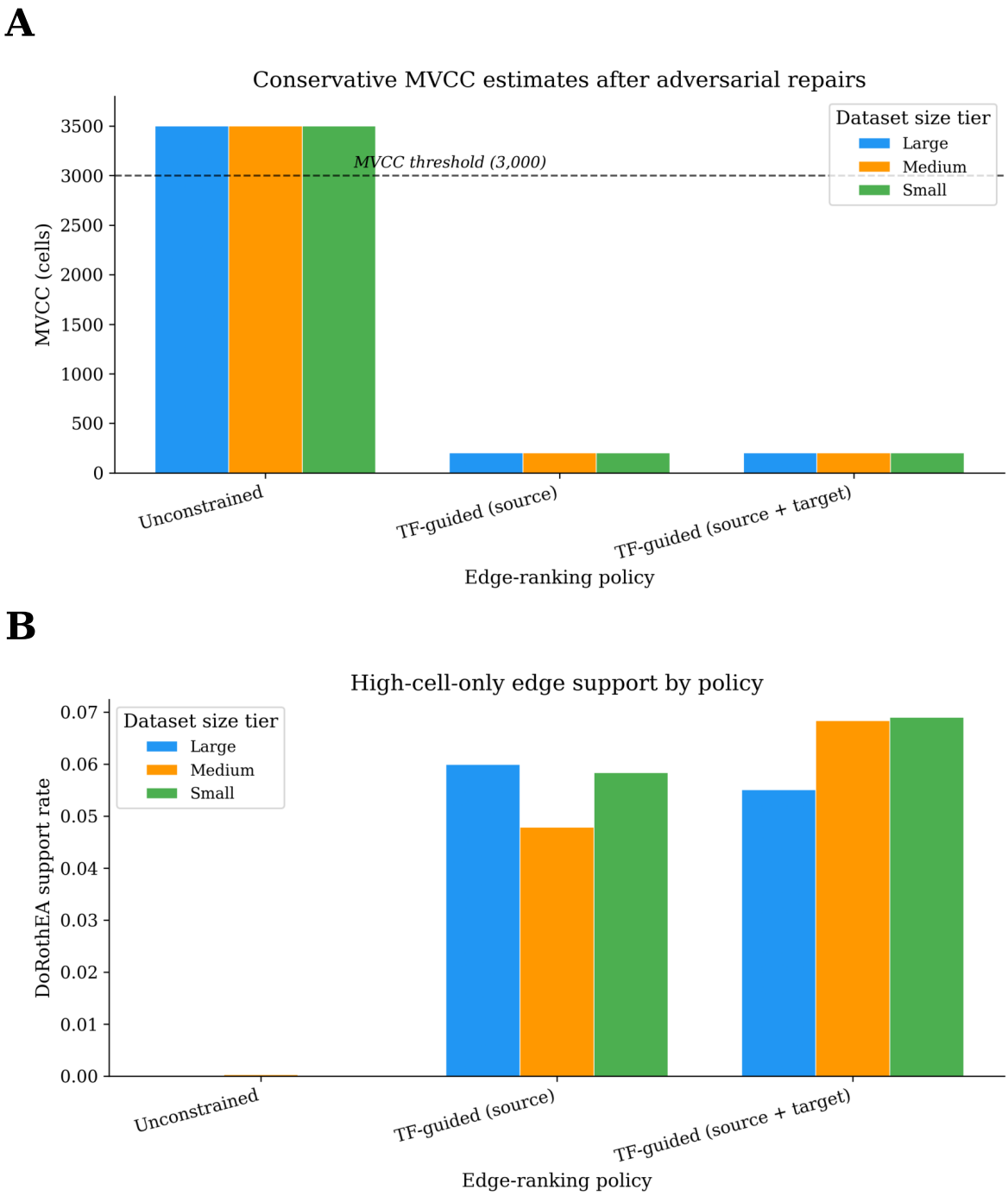

Figure 1: **Cell-count subsampling robustness.** (A) Unconstrained top-1,000 reference Jaccard as a function of cell count across three kidney size tiers; MVCC threshold (>3,000) indicated by dashed line. (B) Prior-constrained versus random-null stability and biological support at 200 and 1,000 cells (64-draw ensemble). Error bars show ensemble interquartile range.

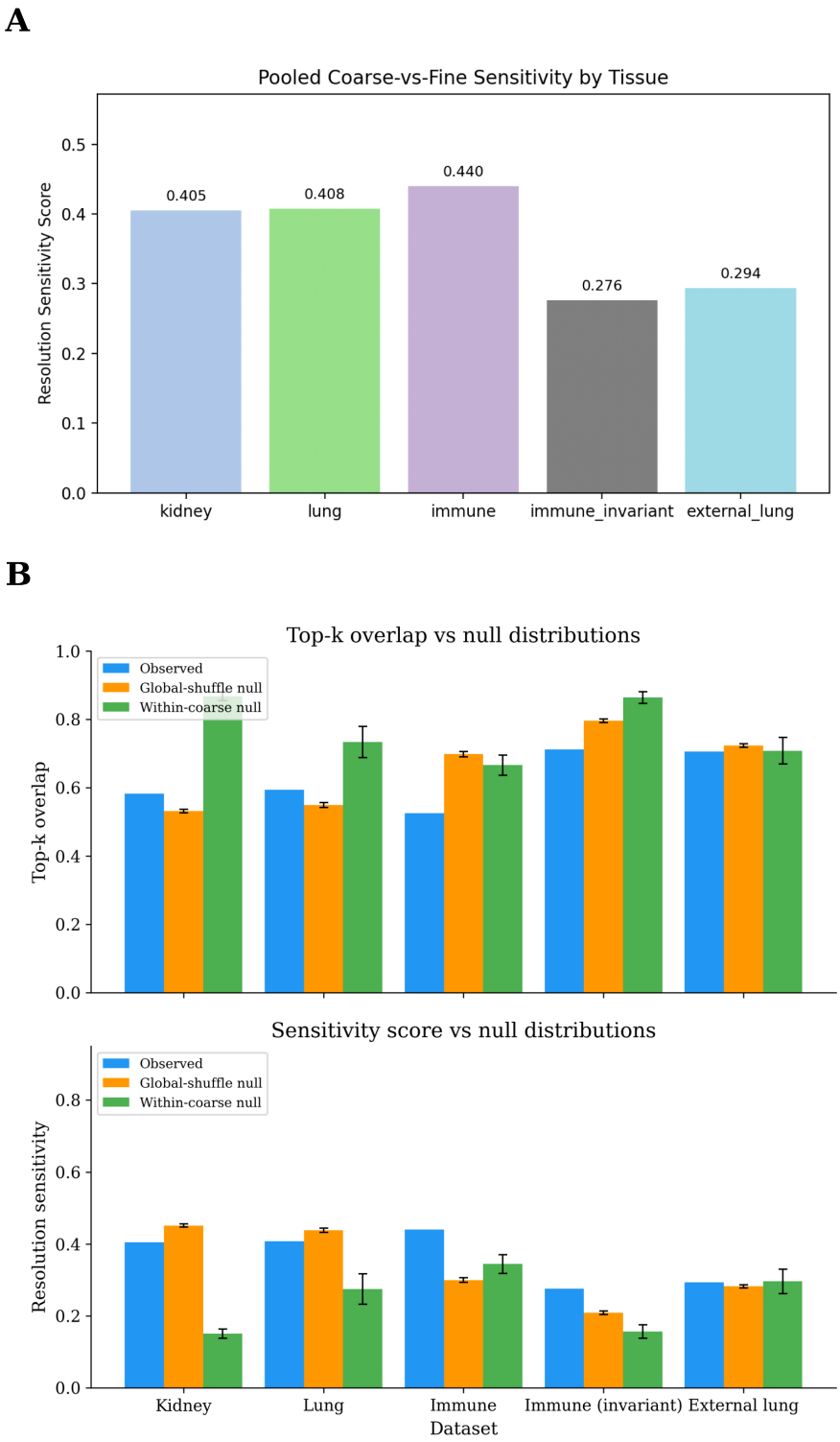

Figure 2: **Cluster-resolution sensitivity and dual-null calibration.** (A) Resolution Sensitivity Score across five cohorts with base recommendation thresholds. (B) Dual-null calibration results: global-shuffle and within-coarse constrained-shuffle pass/fail for each cohort, with final calibrated recommendation.

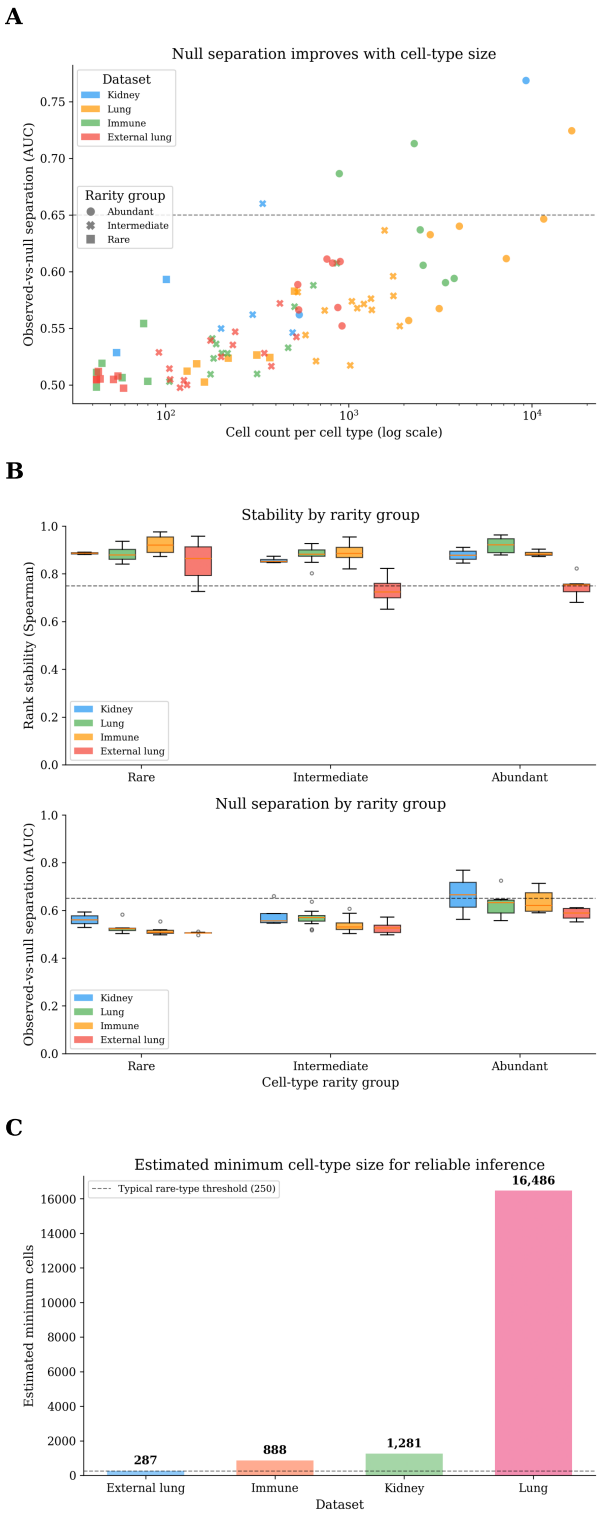

Figure 3: **Rare-cell-type reliability stress test.** (A) Null-separation AUC and top-k Jaccard as functions of cell count, colored by rarity group. (B) Reliability-gate pass rates: 0% rare versus 18% abundant. (C) Minimum reliable cell-size estimates by dataset.

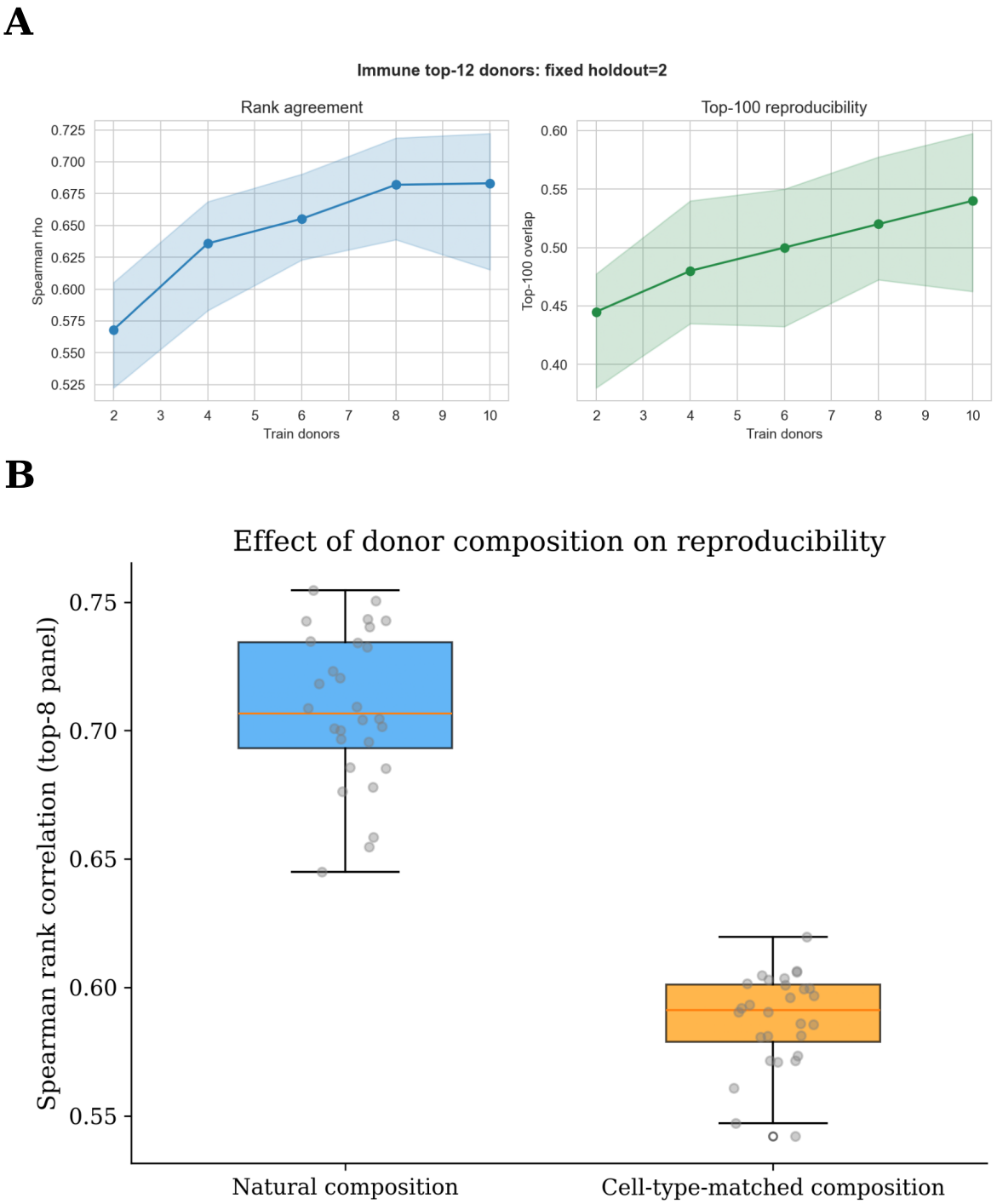

Figure 4: **Donor generalization under fixed-holdout and composition control.** (A) Immune transfer curves under fixed-holdout (holdout= 2) for top-12 and top-20 donor panels. (B) Composition-matched versus unmatched transfer in the top-8 panel. Error bands show interquartile range over 30–40 sampled splits.

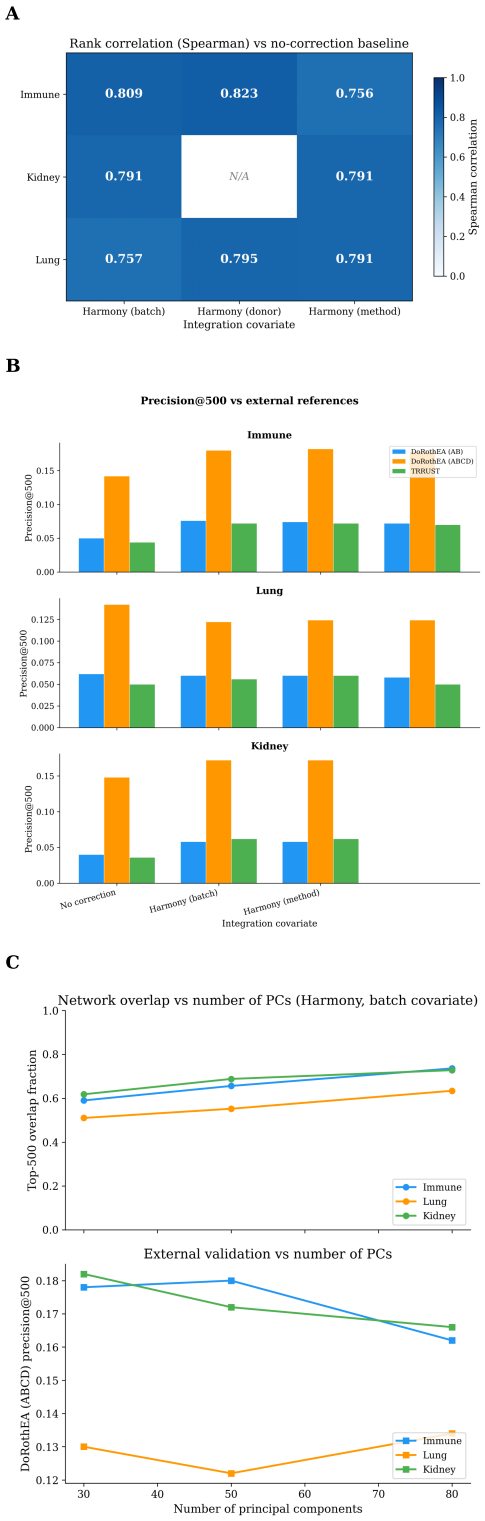

Figure 5: **Integration-method sensitivity.** (A) Rank Spearman and top-500 overlap for Harmony variants across three tissues. (B) Reference precision@500 deltas with paired swap significance; filled bars indicate FDR < 0.05. (C) Harmony hyperparameter sweep: overlap versus DoRothEA precision as a function of n_{pcs}.

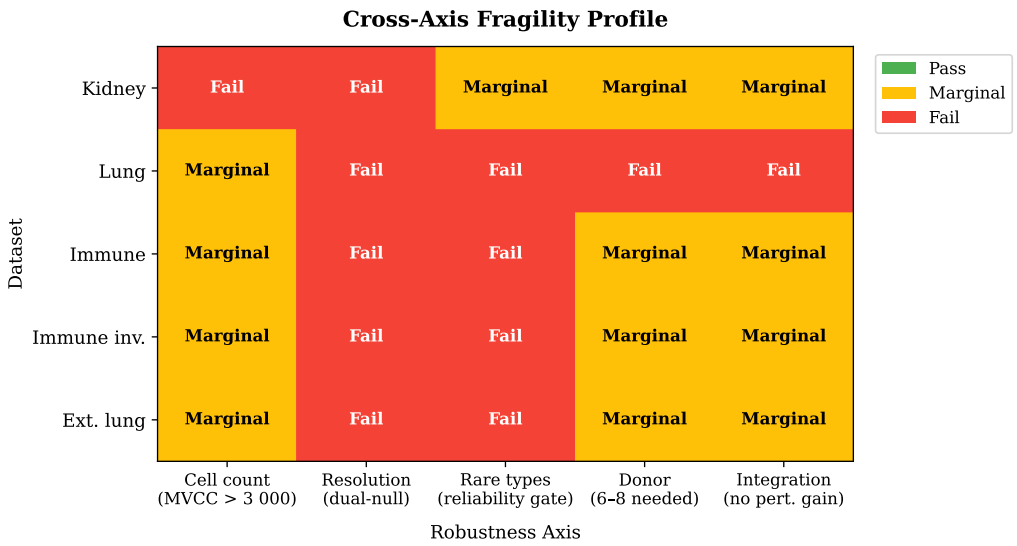

Figure 6: **Cross-axis fragility summary.** Heatmap showing pass (green), marginal (yellow), and fail (red) status for each axis–dataset combination under calibrated criteria.